FIKR Model — Lesson Plan

Learning Leading to Production

Title	Geometric Sequences and Series	**Subject / Unit**	Topic 6 — Exponential and Logarithmic Functions · Lesson 6-7
Grade	Grade 11 — Algebra II	**Date**	Week 8 · Semester 1, 2026–27
Format	[ ] 40-minute [X] 60-minute	**Teacher**	Mr Malek Thiab

Essential Question (curriculum map)

How can we use geometric sequences to model real-world situations?

Lesson Objectives (curriculum map — verbatim)

1. Identify geometric sequences.
2. Write recursive and explicit formulas for geometric sequences.
3. Identify geometric series.
4. Derive the formula for geometric series.
5. Use the formula for geometric series to find sums.

Vocabulary (curriculum map) · Assessments (curriculum map)

Vocabulary: common ratio ; ending index ; geometric sequence ; geometric series ; Index ; sequence of partial sums ; sigma notation ; starting index ; summand ; terms.

Assessments: “Give it a go” questions during practice ; “Time to Check” at the Mastery Gate.

0 Preparation Before Class

Learning Outcome: What will the student be able to do?

Students will identify geometric sequences by testing for a common ratio, and recognise that a sequence is a function whose domain is the positive integers (objective 1; HSF.IF.A.3, HSF.LE.A.2).

Students will write recursive and explicit formulas for geometric sequences and translate between the two forms (objective 2; HSF.BF.A.2).

Students will identify geometric series and read and write sigma notation, naming the starting index, ending index and summand (objective 3).

Students will derive the formula for geometric series (objective 4) and use it to find sums (objective 5), stating that it needs the common ratio not to equal 1 (HSA.SSE.B.4).

Students will use geometric sequences to model real-world situations (essential question; MP.4 and MP.7).

If time runs short, cut Smart Production and carry it into the next session — never the Mastery Gate, which is the evidence.

Pre-Class Preparation — Topic/Link and Evidence of Readiness: What will the student complete before class?

Flipped video (4 min): "Multiply, don't add" — the same number added every time versus the same number multiplied every time. Posted on the LMS 48 hours ahead.

Two embedded EdPuzzle questions: find the common difference of $7,\ 11,\ 15,\ \cdots$, and evaluate $5 \cdot 3^4$.

Evidence of readiness: EdPuzzle completion report plus both embedded answers submitted before the bell.

1 Diagnostic 6 min

Tool Used: What evidence indicates differences in students' current levels?

3-question Quizizz (≤ 4 minutes) — slide 4: evaluate $5 \cdot 3^4$; simplify $\frac{1-2^6}{1-2}$; find a_{10} for $7,\ 11,\ 15,\ \cdots$. No calculator.

Cross-referenced against the EdPuzzle watch analytics.

Answers: 405 ; 63 ; 43 — the first tests a power before a product, the second is the exact shape of the sum formula, the third separates adding from multiplying.

Where is the learning gap? What will you adjust in your instruction accordingly?

Expected gap 1 — applying the exponent to the whole product ($(5 \cdot 3)^4$ for $5 \cdot 3^4$); if Q1 fails, the explicit formula will fail at block 2.

Expected gap 2 — sign slips in $\frac{1-2^6}{1-2}$, which is the denominator and numerator of the sum formula.

Routing: 0–1 correct → re-teach block 1 with the teacher. 2 → straight to the examples. 3 → begin the Investigate prompt immediately.

2 Targeted Instruction 18 min

Required Instruction and Modeling: What needs direct explanation? Which example will demonstrate the thinking process and the expected quality standard?

Slide 5 — Identify a geometric sequence (objective 1; HSF.IF.A.3, HSF.LE.A.2). Divide each term by the one before: $81,\ -27,\ 9,\ -3$ has $r = -\frac{1}{3}$, so it is geometric; $2,\ 4,\ 8,\ 14$ is not ($\frac{4}{2} = 2$ but $\frac{14}{8} = \frac{7}{4}$). Its graph is separate points, not a curve.

Slide 6 — Recursive and explicit formulas (objective 2; HSF.BF.A.2). Recursive: $a_n = r\,a_{n-1}$ with a_1 given. Explicit: $a_n = a_1\,r^{n-1}$. For $5,\ 15,\ 45,\ \cdots$ this gives $a_8 = 5 \cdot 3^7 = 10\,935$. Find r from two terms by dividing and taking a root.

Slide 7 — Geometric series and sigma notation (objectives 3 and 4). A geometric series adds the terms; $\sum_{k=1}^{n} a_1 r^{k-1}$ names the starting index $k = 1$, the ending index n and the summand. Derive: multiply S_n by r, subtract, and the middle terms cancel, leaving $S_n = \frac{a_1(1-r^n)}{1-r}$ for $r \neq 1$.

Slide 8 — Use the formula to find sums (objective 5; HSA.SSE.B.4). Name a_1, r and n first: $\sum_{k=1}^{8} 5 \cdot 3^{k-1} = 16\,400$. When $r = 1$ the formula is undefined and $S_n = n\,a_1$. Partial sums are the sequence of partial sums $S_1,\ S_2,\ \cdots$.

Quick Check for Understanding: What question or evidence confirms that students are ready to move on to practice?

On whiteboards, 60 seconds: for $a_1 = 2,\ r = 5$, find the fourth term.

Expected: $a_4 = 2 \cdot 5^3 = 250$. Watch for $a_4 = 2 \cdot 5^4 = 1250$ (power n instead of $n-1$) and $a_4 = 2 + 15 = 17$ (adding).

80% correct → release guided practice. Below 80% → one more modelled example with a different ratio.

3 Practice 12 min

Guided Task: How will the student progress from guided to independent practice? What feedback will the student use?

Two "Give it a go" problems, identical for every student, worked together with the teacher:

(1) Find the eighth term of $3,\ 6,\ 12,\ \cdots$. (2) Find the sum of the first 5 terms of $1 + 4 + 16 + \cdots$.

Answers: (1) $r = 2$, so $a_8 = 3 \cdot 2^7 = 384$. (2) $a_1 = 1,\ r = 4,\ n = 5$, so $S_5 = \frac{4^5 - 1}{4 - 1} = 341$; check by adding $1 + 4 + 16 + 64 + 256$.

Feedback: worked live on the board step by step; students self-mark and circle their OWN first error.

Independent Task and Feedback Tool: students choose their route and may switch at any time

Delivered through the "Choose Your Route" sheet with live feedback in Pear Deck. Routes are named for the task, not for the student.

PRACTICE — Secure the method. Finding a common ratio, a term from an explicit formula, a recursive and explicit pair, and two sums written in sigma notation.

APPLY — Use it in context. Two terms given and both possible sequences found, a model of a Jeddah electric-bus programme (model values only) with a term and a total, and a sum with a negative ratio.

INVESTIGATE — Find out why. An error analysis of a sum, the partial sums of a series with ratio one third, and the derivation of the sum formula for four terms. "Done when..." criteria are on the sheet.

Every student sits the same Mastery Gate. The gate is not differentiated, because it is the evidence.

4 Production — Stage One 12 min

Production Task: What new task, situation, or product will allow the student to demonstrate learning?

Non-routine transfer task — this context has not appeared in practice (slide 10).

A model Jeddah greening programme (Vision 2030; model values, not real data): $a_1 = 256,\ r = 1.5$ trees planted in month n, and each sapling costs a model 25 SAR.

(1) Write the recursive and explicit formulas and find month 6. (2) Use the sum formula for the first 6 months, and find the cost in SAR. (3) Find the first month in which the running total passes 10 000 trees.

Answers: (1) $a_1 = 256,\ a_n = 1.5\,a_{n-1}$ and $a_n = 256(1.5)^{n-1}$, so $a_6 = 256 \times 7.59375 = 1944$ trees. (2) $S_6 = \frac{256(1-1.5^6)}{1-1.5} = 5320$ trees, costing $5320 \times 25 = 133\,000$ SAR. (3) $S_n = 512\,(1.5^n - 1)$; $S_7 = 8236$ is

below 10 000 and $S_8 = 12\,610$ is above, so month 8.

Students then hand their reasoning to the AI critic and ask it to challenge any step they have not justified.

Product Quality Criteria and the Role of AI: What is the minimum acceptable standard? What must the student complete independently?

Minimum acceptable standard: a_1, r and n named before the formula is used, exact values before any rounding, and units (trees, SAR) in every answer.

Role of AI: critic only. Briefed prompt — "Challenge my reasoning and point out any step I have not justified." Never "solve this".

Completed independently by the student: the written justification of each step. AI may challenge it; it may not write it.

5 Presentation — Evidence of Learning 6 min

Evidence of Learning and Success Criterion: What will the student present or explain, and how? What must be clearly demonstrated for learning to be considered achieved?

"Time to Check" — one individual task. No notes, no partner, no AI.

1. Find the next term of $96,\ -48,\ 24,\ \cdots$. 2. For $a_1 = 7,\ a_n = 2a_{n-1}$, find a_6. 3. Evaluate $\sum_{k=1}^{5} 4 \cdot 3^{k-1}$.

Answers: 1. $r = -\frac{1}{2}$, next term -12. 2. $a_n = 7 \cdot 2^{n-1}$, so $a_6 = 224$. 3. $a_1 = 4,\ r = 3,\ n = 5$, so $S_5 = 484$.

Done when: the ratio is found before the next term, the explicit formula is written before a_6, and the sum names a_1, r and n.

Scored live in Pear Deck as submissions arrive; result logged in the Impact Dashboard Mastery Rate.

6 Final Product — Refinement 6 min

Refinement and Final Product: What revision will the student make based on feedback? What will the final product look like?

Students revise using the AI critique and the teacher's micro-conference note. Final product, three equally acceptable formats: a Canva one-pager, a photographed hand-written solution, or a 45-second voice note.

Required content: the greening-programme solution with a_1, r and n named at each step; ONE worked non-example — using r^n for the explicit term, or using the sum formula with $r = 1$ — with a sentence on what goes wrong; and a caption on how a planner would use the totals.

Graded on originality and clarity of reasoning, not correctness alone.

Presentation/Publication and Reflection: Where will the product be presented or published? What reflection question will close the learning cycle?

Published to the class LMS portfolio; the strongest go on the Mathematics Department wall.

Reflection: "How do you tell an arithmetic sequence from a geometric one?"

Total minutes across the seven stages: **60** (6 + 18 + 12 + 12 + 6 + 6) — matches the 60-minute format, including transitions.

★ Required Pillars

Differentiation · Adaptive Learning · Saudi Connection · Exam Alignment

Differentiation — three routes named for the task; students choose, and may switch at any time

PRACTICE — Secure the method. A common ratio, a term from an explicit formula, a recursive and explicit pair, and two sums. Done when: the ratio is found by dividing consecutive terms, and each formula reproduces the first terms.

APPLY — Use it in context. Two terms given, a bus-programme model, and a sum with a negative ratio. Done when: the number of steps between terms is shown, and each sum names its first term, ratio and number of terms.

INVESTIGATE — Find out why. An error analysis, partial sums that approach a value, and the derivation of the sum formula.

No student is labelled. The routes describe tasks; every student sits the same Mastery Gate.

Adaptive Learning — how the lesson branches on evidence

Entry branch — the Quizizz result routes each student to the re-teach table, straight to the new content, or the Investigate prompt.

Mid-lesson branch — Pear Deck flags error patterns during independent practice; flagged students get a 2-minute micro-conference while others continue.

Exit branch — the Mastery Gate routes each student: PASS → Enrichment & Challenge; NOT YET → Targeted Learning Clinic at the start of the next lesson.

Routing is communicated privately through the LMS, never read aloud.

Saudi Culture & Real-World Connection

A Jeddah greening programme (Vision 2030; model values only); the A3 project compares a faster and a slower neighbourhood.

SAT / SAAT (Tahsili) / GAT (Qudurat) Connection

SAT — Advanced Math: first term 2, each term 3 times the one before; sum of the first 5 terms. $S_5 = \frac{2(3^5-1)}{3-1} = 242$. Name the trap: choosing 162, the fifth term, as a partial step.

GAT (Qudurat) — quantitative comparison: A is the 6th term of $1,\ 2,\ 4,\ \cdots$; B is the sum of its first 5 terms. $A = 32$ and $B = 31$, so A is greater. Name the trap: taking the 6th term as 2^6.

SAAT (Tahsili) — $a_3 = 12,\ a_6 = 96$: $r = 2$, $a_1 = 3$, $S_5 = 93$. Name the trap: counting terms, not steps.

Suggestions for Improvement: